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Data Scientist
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Summary

Data Scientist with hands-on experience in **Python, SQL, Machine Learning, and Predictive Modeling**, building and independently validating models for classification, regression, anomaly detection, and forecasting. Skilled in data analysis, feature engineering, and statistical modeling, with a consistent record of auditing and correcting flawed models – finding leakage, governance violations, and evaluation bugs – before they reach a business decision. Builds **GenAI/LLM applications** with real evaluation harnesses. Strong communicator, translating technical findings clearly to both technical and business stakeholders.

Experience

Data Scientist Sep 2025 – Dec 2025
Upgrad School of Technology
Promoted from Data Science Content Strategist Associate following upGrad's acquisition of Aimerz.ai – continued with the same team.

- Designed and executed **SQL and Python-based analytical workflows** to extract, clean, and engineer features across structured and unstructured datasets, assessing **data quality and integrity** to ensure reliable, reproducible outputs.
- Performed **anomaly detection** using the Interquartile Range (IQR) method during data preprocessing, followed by trend analysis and correlation studies to identify key business drivers and validate model performance.
- Developed **KPI dashboards** and business performance reports using Python, SQL, Pandas, NumPy, Matplotlib, and Seaborn, enabling stakeholder visibility into key metrics driving business decisions.
- Built and tested **machine learning and predictive analytics workflows** using Python, SQL, Pandas, and Scikit-learn, including data preprocessing and feature engineering.
- Partnered with internal and cross-functional business stakeholders, gathering requirements and structuring analysis to solve real-world business problems.

Data Science Content Strategist Associate Aug 2024 – Sep 2025
Aimerz.ai (acquired by upGrad in 2025)

- Joined as the **first member of Aimerz's Data Science team**, designing and developing **15+ Python-based analytical workflows, predictive modelling projects, and machine learning solutions** in Google Colab.
- Built and improved analytical workflows using Python and SQL, automating data cleaning, exploratory data analysis (EDA), and feature engineering.
- Applied **regression, classification, and statistical modelling** techniques for predictive modelling, validating model performance.
- Conducted **model evaluation**, data validation, and technical reporting across 15+ Python-based projects.

Projects

Credit Default Risk Model – Independent Validation & Fair-Lending Remediation
GitHub: melvinmathew9991/credit-default-risk-model-audit

- Took delivery of a LightGBM credit-default model reporting **0.9636 AUC**, reproduced it to 15 decimal places, then proved by counterfactual that **85% of its performance was leakage** – five features describing repayment on the loan being scored.
- Rebuilt under a **governance policy enforced in code** – excluding gender, marital status, and pincode as prohibited bases under **ECOA/Reg B** – with the remediated model scoring a defensible **Gini 0.2914** on a hold-out touched exactly once.
- Measured the cost of each governance decision separately: fair-lending compliance was free (Gini changed by +0.0013), while leakage removal accounted for 96% of the total performance drop.
- Shipped as a decision system: **isotonic calibration**, **SHAP-based** adverse-action reason codes, a WOE logistic-regression challenger, production **drift monitoring (PSI)**, backed by 243 automated tests and CI.

AML Fraud Detection Pipeline (DuckDB, Optuna, MLflow, FastAPI, Docker, CI/CD)
GitHub: melvinmathew9991/aml-fraud-detection-pipeline

- Built an end-to-end **fraud/anomaly-detection system** (Python, **DuckDB**, Scikit-learn) on **6.36 million real** PaySim mobile-money transactions (0.13% fraud rate), running **3-fold time-based cross-validation** across 8 models with **Optuna**-based hyperparameter optimization and **MLflow** tracking – best model (LightGBM) reaching **PR-AUC 0.997**.
- Replaced label-derived evaluation with a **capacity-based operating point** to set a deployable threshold before ground truth exists – **250 reviews/day yields 95.1% recall at zero false positives**.
- Diagnosed and fixed a **production-stability bug** crashing an 8GB machine on full-dataset runs, rewriting the aggregation as a **DuckDB SQL window function** and cutting peak memory from a multi-GB spike to **1.4GB**.
- Backed the system with a **182-test automated suite** and shipped a **FastAPI** scoring service and **Docker**-containerized image behind a green **3-job CI/CD** gate.

MEDBOT – Knowledge-Grounded Medical RAG Assistant (Gen AI / LLM)
GitHub: melvinmathew9991/Medical_RAG_Chatbot

- Designed and built a **GenAI/RAG application** (Python, **LangChain**, **Google Gemini**, local **fastembed** embeddings, **FAISS** – 1,225/1,225 indexed chunks) for knowledge-grounded medical question-answering, migrating off a paid OpenAI dependency onto a fully free-tier stack.

- Fixed **three real bugs** surfaced by end-to-end testing: a PubMed TLS interception issue, a Wikipedia 403 error, and a SerpAPI exception-handling bug where a bad API key silently discarded all three external sources instead of just the failing one.
- Applied and evaluated **Chain-of-Thought prompting** via a structured **A/B test with an ablation study**, alongside a 27-example **few-shot prompt template** with semantic-similarity example selection.
- Built a genuine **evaluation harness** – not qualitative spot-checks – measuring retrieval **Precision@4 (0.83)** and generation **groundedness (0.83)** against a 24-question test set, using it to discover a corpus-coverage gap and a reproducible false-refusal bug invisible to code review.
- Shipped behind a **CI/CD** pipeline with an index-integrity gate, deployed via **Streamlit** in a containerized (Dev Containers) reproducible environment.

Skills

Languages & Tools: Python, R, SQL, SQLite, DuckDB, FastAPI, Advanced Excel, Git/GitHub, Streamlit, Docker, Jupyter
Machine Learning & Predictive Modelling: Supervised Learning (Logistic Regression, Random Forest, XGBoost, LightGBM), Classification, Regression, Anomaly Detection, Clustering (KMeans), Hyperparameter Optimization (Optuna, GridSearchCV), Time-Series Forecasting, Cross-Validation
Model Governance & Responsible AI: Fairness Auditing, Fair-Lending Testing (ECOA/Reg B awareness), Model Calibration (Isotonic/Platt), Explainability (SHAP), Production Drift Monitoring (PSI), Leakage Detection
Gen AI / LLM & Prompt Engineering: Generative AI, RAG Architecture, Google Gemini, LangChain, FAISS (Vector Database), LLM Evaluation
Data Analysis & Statistics: Hypothesis / Significance Testing, Statistical Modelling, Statistical Inference, EDA, Data Verification & Audit, Quantitative Analysis
Model Lifecycle, MLOps & Deployment: End-to-End Model Development, Model Deployment (FastAPI, Streamlit), Containerization (Docker), Experiment Tracking (MLflow), CI/CD Pipelines (GitHub Actions), Automated Regression Testing, Reproducibility
Databases & Data Engineering: MySQL, PostgreSQL, SQLite, DuckDB, SQL Query Writing, Multi-table Joins & CTEs, ETL Workflows, Graph Analytics (NetworkX), Data Modelling
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras, Statsmodels, SciPy, Matplotlib, Seaborn
Visualization & Collaboration: Tableau, Power BI, Dashboard Development, Data Storytelling, Stakeholder Management, Cross-functional Collaboration, Requirements Gathering, Agile/Scrum

Education

BSc Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore — 2017 – 2020
Data Science Bootcamp, Brototype, Bengaluru — 2023 – 2024